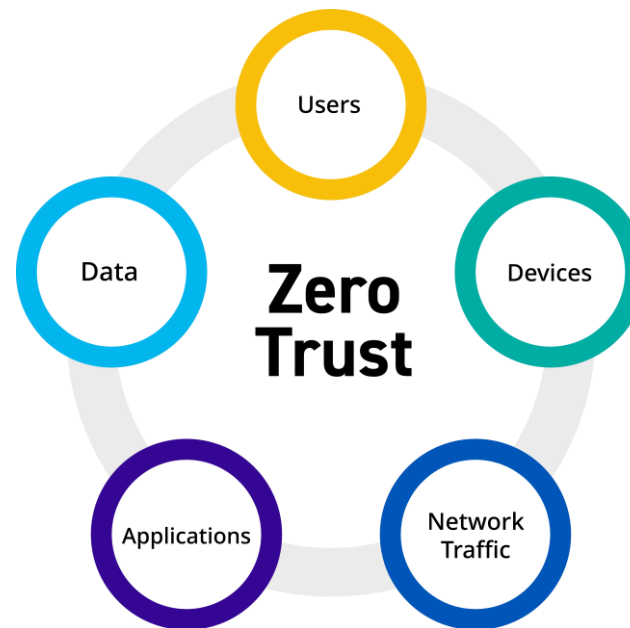




Cyber Security Awareness Program
Research Project
Zero Trust Identity Framework for Cloud Security

Introduction

- **Zero Trust** redefines this approach by assuming **Trust No One — Always Verify**. It continuously verifies every user, device, and access request, regardless of location.
- **Focus of the Research:**
This project explores how **Zero Trust principles** can strengthen **Identity and Access Management (IAM)** in **cloud environments** to protect against data breaches and unauthorized access.



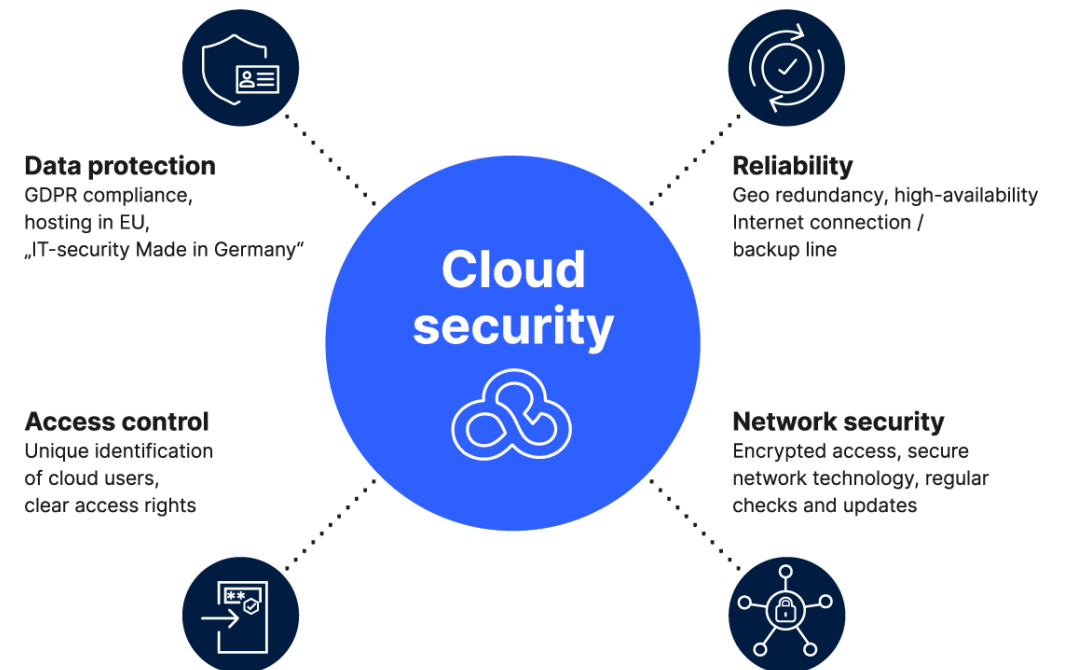
Cloud Security

Cloud Security involves protecting data, applications, and infrastructure hosted in the cloud from threats such as:

- Unauthorized access
- Misconfiguration
- Data leaks
- Insider attacks

It follows a **shared responsibility model**:

- **Cloud Provider:** Secures the infrastructure
- **Customer:** Secures access, data, and identities



Zero Trust Security

Zero Trust is a cybersecurity framework built on the principle of **Never Trust, Always Verify**.

It means:

- No user or device is trusted automatically — inside or outside the network.
- Every access request is verified continuously using strong authentication, authorization, and monitoring.

Core Principles

1. **Verify Explicitly** – Authenticate and authorize based on all data points (identity, device, location).
2. **Use Least Privilege Access** – Give users the minimal level of access needed.
3. **Assume Breach** – Always monitor and prepare as if the network is already compromised.

Identity as the Foundation of Zero Trust

In cloud environments, **identity is the new perimeter** — not firewalls.

Identity controls include:

- **Authentication:** Passwords, Biometrics, Multi-Factor Authentication (MFA)
- **Authorization:** Role-Based Access Control (RBAC), Policy-Based Access
- **Identity Federation:** Single Sign-On (SSO) across cloud services
- **Privileged Access Management (PAM):** Protects admin and high-level accounts

Example:

When a user logs into a cloud service (like Microsoft Azure), Zero Trust checks:

- User identity
- Device health
- Network location
- Risk score (based on previous behavior)

Only if all checks pass, access is granted.

Practical Zero Trust Habits for Everyday Users

Zero Trust for Users (Identity & Access Protection)

Identity Protection

- Use strong, unique passwords for each account.
- Use a **password manager** (Bit warden, 1Password, etc.).
- Enable **MFA or biometric authentication**.
- Never share login details — even with IT support.

Cloud & Data Access

- Check **who has permission** before sharing files (Google Drive, OneDrive, Azure).
- Avoid logging in from **public/shared devices**.
- Always **sign out properly** from VPN or corporate networks.

💡 *Tip:* Identity is your first line of defense — protect it like your digital fingerprint.

Practical Zero Trust Habits for Everyday Users

Zero Trust for Users (Device & Behaviour Security)

Device Trust

- Keep all devices **patched and updated**.
- Install apps only from **trusted sources**.
- Use **encryption** and **screen lock** on phones and laptops.

Awareness of Phishing & Social Engineering

- Don't click suspicious links or attachments.
- Always verify the sender through another channel.
- Watch for red flags: **misspellings, fake domains, urgency**.

Continuous Verification

- Regularly check **logged-in devices**.
- Sign out old or unused sessions.
- Review app permissions every few months.

Remember: Trust nothing, verify everything — that's the Zero Trust mindset!

Expected Outcomes

- Improved access control and reduced risk of identity theft.
- Increased awareness of Zero Trust principles in cloud environments.
- Framework to guide organizations transitioning from perimeter-based to Zero Trust models.
- Better compliance with standards like ISO 27001, GDPR, and NIST 800-207.

Conclusion

Zero Trust is not a product — it's a **security mindset**.

By focusing on identity, continuous verification, and least privilege, organizations can secure their cloud environments against modern cyber threats.

In a Zero Trust world, identity is the new firewall — protect it relentlessly.

Awareness Message

- Use strong, unique passwords
- Enable multi-factor authentication
- Be cautious with links and emails
- Keep systems and devices updated
- Remember: Security starts with you!



THANK YOU

